# MAE 6246 - Linear Systems and Control

Week 12: Probability and Gaussian Parameter Estimation

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## Contents


**1 Probability foundations** **2**

**2 Mean, covariance, and Gaussian distributions** **3**

**3 Transformations, joint distributions, and conditioning** **4**

**4 Gaussian parameter estimation from a noisy measurement** **5**

**5 Estimation error and posterior covariance** **6**

**6 Orthogonality: no predictable error remains** **7**

**7 MAP and MMSE interpretations** **8**

**8 Example: the position of a one-dimensional vehicle** **9**

**9 Summary, practice, and the next lecture** **10**


**Main question:** How do we combine a prior belief and a noisy measurement to estimate an unknown quantity and quantify the remaining uncertainty?

**Lecture scope:** probability concepts; mean and covariance; Gaussian distributions and conditioning; static parameter estimation; unbiasedness, covariance, and orthogonality; maximum a posteriori (MAP) and minimum mean-square error (MMSE) estimates.

**Notation:** following the handwritten lecture, the prior is $x \sim \mathcal{N}(\bar{x}, M)$ and the measurement is $z = Hx+v$, with $v \sim \mathcal{N}(0, R)$. Throughout this note, the estimation error is $e = \hat{x} - x$.

**Next lecture:** Week 13 develops the Kalman filter for a dynamic state and combines it with LQR to obtain LQG control. This week's Gaussian measurement update is the foundation for that recursion.

# 1 Probability foundations

## 1.1 From state observers to estimation under uncertainty

State feedback uses $u = -K_{\mathrm{LQR}}x$, but a sensor may reveal only part of the state. A deterministic observer uses the discrepancy between the measured and predicted output:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}).$$

Pole placement chooses $L$ to obtain desired error dynamics. To account for uncertainty in the state and sensor, we need a probability model. This lecture first studies one unknown parameter and one measurement. The dynamic Kalman filter is the next step.

## 1.2 Experiments, outcomes, events, and random variables

- An *experiment* produces one of several possible outcomes, such as rolling a die.
- An *outcome* is one particular result, such as rolling a 4.
- The *sample space* $\Omega$ contains all possible outcomes. For a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- An *event* is a subset of $\Omega$. The event of an even result is $E = \{2, 4, 6\}$.
- A probability assigns a number in $[0, 1]$ to an event. For a fair die, $\mathbb{P}(E) = 1/2$.
- A *random variable* maps an outcome to a numerical value. For two dice, $X$ could be their sum and $Z$ the first die's value.

The random variable $X$ describes the experiment before the outcome is known; $x$ denotes a realized value. We also use lowercase $x$ for a random state vector when its meaning is clear from context.

## 1.3 Probability mass and probability density

For a discrete random variable, probabilities are assigned to possible values. For a continuous random variable with density $p_X$,

$$\mathbb{P}(a \le X \le b) = \int_a^b p_X(x)\,dx, \qquad p_X(x) \ge 0, \qquad \int_{-\infty}^{\infty} p_X(x)\,dx = 1.$$

Probability is the *area* under the density over an interval. A density value is not itself a probability and can exceed one; for a continuous density, $\mathbb{P}(X = x) = 0$ at any one point.

## 1.4 Conditioning means incorporating known information

For events $E, F$ with $\mathbb{P}(F) > 0$, $\mathbb{P}(E \mid F) = \mathbb{P}(E \cap F)/\mathbb{P}(F)$. If $X$ is the sum of two independent fair dice and $Z$ is the first die,

$$\mathbb{P}(X = 2 \mid Z = 3) = 0, \qquad \mathbb{P}(X = 2 \mid Z = 1) = \frac{1}{6}.$$

Knowing the first die changes the possible sums. In estimation, a sensor reading plays the role of this new information. For continuous variables, conditional densities are defined through the joint density:

$$p_{X|Z}(x \mid z) = \frac{p_{X,Z}(x, z)}{p_Z(z)}, \qquad p_Z(z) > 0.$$

# 2 Mean, covariance, and Gaussian distributions

## 2.1 Mean and variance

The mean is the probability-weighted average:

$$\mu = \mathbb{E}[X] = \int_{-\infty}^{\infty} x p_X(x)\, dx, \qquad \mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2].$$

For a discrete random variable, replace the integral by a sum. A fair die has $\mathbb{E}[X] = (1+\cdots+6)/6 = 3.5$; the mean need not be a possible outcome.

For $x \in \mathbb{R}^n$, define

$$\mu = \mathbb{E}[x], \qquad \Sigma = \mathbb{E}[(x-\mu)(x-\mu)^T].$$

The covariance is symmetric and positive semidefinite because

$$a^T \Sigma a = \mathbb{E}[(a^T(x-\mu))^2] \geq 0.$$

Its diagonal entries are variances; off-diagonal entries are covariances. Standard deviations are $\sqrt{\Sigma_{ii}}$, with the same units as the associated state components. Also, $\mathbb{E}[\|x-\mu\|^2] = \operatorname{tr}\Sigma$.

## 2.2 The scalar Gaussian distribution

A Gaussian random variable $X \sim \mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$, has density

$$\boxed{p_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].}$$

The mean $\mu$ sets the center; the standard deviation $\sigma$ sets the width. The second argument of $\mathcal{N}$ is the *variance*, not the standard deviation.

| Interval | Gaussian probability |
|---|---|
| $\mu - \sigma \leq X \leq \mu + \sigma$ | approximately 68.27% |
| $\mu - 2\sigma \leq X \leq \mu + 2\sigma$ | approximately 95.45% |
| $\mu - 3\sigma \leq X \leq \mu + 3\sigma$ | approximately 99.73% |

As an illustrative height model, take $\mu = 69$ inches and $\sigma = 3$ inches. Then $\mathbb{P}(66 \leq X \leq 72) \simeq 0.6827$ and $\mathbb{P}(X \leq 75) \simeq 0.9772$.

## 2.3 Why Gaussian models arise; the vector density

The central limit theorem explains why sums of many small independent effects often look approximately Gaussian. Precisely, for independent identically distributed variables with finite mean $\mu$ and nonzero finite variance $\sigma^2$, the *normalized* sum $(\sum_{i=1}^N X_i - N\mu)/(\sigma\sqrt{N})$ converges in distribution to $\mathcal{N}(0,1)$.

For $x \sim \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$,

$$p(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \exp\left[-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right].$$

A Gaussian with singular covariance is still meaningful, but does not have this full-dimensional density.

# 3 Transformations, joint distributions, and conditioning

## 3.1 Linear transformations preserve Gaussianity

If $x \sim \mathcal{N}(\mu, \Sigma)$ and $A, b$ are deterministic, then

$$\boxed{Ax + b \sim \mathcal{N}(A\mu + b, A\Sigma A^T).}$$

The mean and covariance formulas themselves hold for any distribution with finite second moments. Gaussianity additionally determines the entire transformed distribution from those two quantities.

For random vectors $x, y$, the cross-covariance is

$$\Sigma_{xy} = \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])^T], \qquad \Sigma_{yx} = \Sigma_{xy}^T.$$

Independence implies zero cross-covariance. Zero cross-covariance implies independence for *jointly Gaussian* vectors; this implication does not hold in general.

## 3.2 Marginal and conditional Gaussian distributions

Suppose

$$\begin{bmatrix} x \\ z \end{bmatrix} \sim \mathcal{N}\left( \begin{bmatrix} \mu_x \\ \mu_z \end{bmatrix}, \begin{bmatrix} \Sigma_{xx} & \Sigma_{xz} \\ \Sigma_{zx} & \Sigma_{zz} \end{bmatrix} \right), \qquad \Sigma_{zz} \succ 0.$$

The marginal distributions are $x \sim \mathcal{N}(\mu_x, \Sigma_{xx})$ and $z \sim \mathcal{N}(\mu_z, \Sigma_{zz})$. Once a value of $z$ is known,

$$\boxed{\begin{aligned} \mathbb{E}[x \mid z] &= \mu_x + \Sigma_{xz}\Sigma_{zz}^{-1}(z - \mu_z), \\ \mathrm{Cov}(x \mid z) &= \Sigma_{xx} - \Sigma_{xz}\Sigma_{zz}^{-1}\Sigma_{zx}. \end{aligned}}$$

The difference $z - \mu_z$ measures how surprising the observation is relative to its predicted mean. The cross-covariance determines how that surprise changes the estimate of $x$.

## 3.3 Why the conditioning formula has this form

Define $B = \Sigma_{xz}\Sigma_{zz}^{-1}$ and

$$r = x - \mu_x - B(z - \mu_z).$$

Then $\mathbb{E}[r] = 0$ and

$$\mathrm{Cov}(r, z) = \Sigma_{xz} - B\Sigma_{zz} = 0.$$

Because $(r, z)$ is jointly Gaussian, $r$ is independent of $z$. Its covariance is $\Sigma_{xx} - \Sigma_{xz}\Sigma_{zz}^{-1}\Sigma_{zx}$. Thus, in

$$x = \mu_x + B(z - \mu_z) + r,$$

conditioning on $z$ fixes the first two terms and leaves the distribution of $r$ unchanged. This proves the displayed conditional mean and covariance.

# 4 Gaussian parameter estimation from a noisy measurement

## 4.1 Prior belief and measurement model

We wish to estimate an unknown $x \in \mathbb{R}^n$ using

$$\boxed{x \sim \mathcal{N}(\bar{x}, M), \qquad z = Hx + v, \qquad v \sim \mathcal{N}(0, R).}$$

Here $z \in \mathbb{R}^p$, $H \in \mathbb{R}^{p \times n}$, $M \succeq 0$, and $R \succ 0$. Assume $x$ and $v$ are independent. The prior describes uncertainty before the measurement; $R$ describes sensor-noise covariance. The physical parameter is fixed during the measurement, while our uncertainty about it is represented probabilistically.

## 4.2 Construct the joint distribution

Independence gives

$$\begin{bmatrix} x \\ v \end{bmatrix} \sim \mathcal{N}\left( \begin{bmatrix} \bar{x} \\ 0 \end{bmatrix}, \begin{bmatrix} M & 0 \\ 0 & R \end{bmatrix} \right).$$

Use the transformation

$$\begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} I_n & 0 \\ H & I_p \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$

to obtain

$$\boxed{\begin{bmatrix} x \\ z \end{bmatrix} \sim \mathcal{N}\left( \begin{bmatrix} \bar{x} \\ H\bar{x} \end{bmatrix}, \begin{bmatrix} M & MH^T \\ HM & HMH^T + R \end{bmatrix} \right).}$$

In particular, $\mathbb{E}[z] = H\bar{x}$ and the measurement residual $\nu = z - H\bar{x}$ has covariance $S = HMH^T + R \succ 0$.

## 4.3 Condition on the measurement

Applying the joint Gaussian formula gives

$$\boxed{\begin{aligned} &S = HMH^T + R, \qquad K = MH^T S^{-1}, \\ &\hat{x} = \bar{x} + K(z - H\bar{x}), \quad P = M - MH^T S^{-1} HM. \end{aligned}}$$

Thus $x \mid z \sim \mathcal{N}(\hat{x}, P)$. The estimate is a prior mean plus a weighted measurement residual. The gain $K \in \mathbb{R}^{n \times p}$ converts a sensor discrepancy into a correction of the unknown parameter.

For a scalar position sensor $H = 1$,

$$K = \frac{M}{M + R}, \qquad \hat{x} = (1 - K)\bar{x} + Kz, \qquad P = \frac{MR}{M + R}.$$

A more accurate sensor (smaller $R$) receives more weight. A more certain prior (smaller $M$) receives more weight. These scalar statements do not imply entrywise monotonicity of general matrix gains.

# 5 Estimation error and posterior covariance

## 5.1 A consistent error convention

Throughout this note, $e = \hat{x} - x$. Let $\delta x = x - \bar{x}$. Then

$$\hat{x} = \bar{x} + KH\delta x + Kv, \qquad \boxed{e = -(I - KH)\delta x + Kv.}$$

Using $x - \hat{x}$ instead reverses the error sign but leaves its covariance and zero-orthogonality identities unchanged.

Since $\mathbb{E}[\delta x] = 0$ and $\mathbb{E}[v] = 0$,

$$\boxed{\mathbb{E}[e] = 0.}$$

This is unbiasedness averaged over the assumed prior and measurement noise. It does not claim unbiasedness for every fixed true parameter value. In fact,

$$\mathbb{E}[e \mid x] = (I - KH)(\bar{x} - x),$$

which is generally nonzero. A Bayesian estimate trades reliance on the prior against reliance on the measurement.

## 5.2 Derive the Joseph covariance form

Expand the product $ee^T$ and take its expectation:

$$\begin{aligned}\mathbb{E}[ee^T] = (I - KH)\mathbb{E}[\delta x \delta x^T](I - KH)^T + K\mathbb{E}[vv^T]K^T \\ - (I - KH)\mathbb{E}[\delta x v^T]K^T - K\mathbb{E}[v\delta x^T](I - KH)^T.\end{aligned}$$

The cross terms vanish by independence, so

$$\boxed{P = (I - KH)M(I - KH)^T + KRK^T.}$$

For the optimal gain $K = MH^T S^{-1}$, expand and use $KS = MH^T$:

$$\begin{aligned}P &= M - KHM - MH^T K^T + K(HMH^T + R)K^T \\ &= M - KHM = (I - KH)M.\end{aligned}$$

The two-term Joseph form is useful numerically because each term is positive semidefinite. In this model $P = \mathrm{Cov}(x \mid z)$ is independent of the observed value of $z$, and also equals the unconditional error covariance.

## 5.3 What the covariance says about accuracy

Conditioning reduces uncertainty in the positive-semidefinite ordering:

$$M - P = MH^T S^{-1} HM \succeq 0.$$

For a scalar sensor, $P = MR/(M + R) \leq \min(M, R)$ when $M, R > 0$. This describes expected squared error, not a guarantee that every realized error gets smaller. The estimate can move away from the actual state on one noisy measurement even though its mean-square performance improves.

# 6 Orthogonality: no predictable error remains

## 6.1 The error is uncorrelated with the estimate

Because $\hat{x} = \mathbb{E}[x \mid z]$,

$$\mathbb{E}[e \mid z] = \hat{x} - \mathbb{E}[x \mid z] = 0.$$

Any function $g(z)$ is fixed after conditioning on the measurement. Hence, for square-integrable $g$,

$$\boxed{\mathbb{E}[e\, g(z)^T] = \mathbb{E}\Big[\mathbb{E}[e \mid z] g(z)^T\Big] = 0.}$$

In particular, choose $g(z) = \hat{x}$:

$$\boxed{\mathbb{E}[e\hat{x}^T] = 0.}$$

Since $\mathbb{E}[e] = 0$, this says $\mathrm{Cov}(e, \hat{x}) = 0$. In the linear Gaussian problem, the error and estimate are jointly Gaussian, so they are also independent.

The interpretation is statistical: among repeated experiments producing the same estimate, the errors average to zero, since $\mathbb{E}[e \mid \hat{x}] = 0$. Orthogonality here uses an expectation inner product. It does not mean the realized vectors $e$ and $\hat{x}$ are perpendicular in each experiment, or that the error itself is zero.

## 6.2 What if the scalar cross moment is positive?

For a candidate estimate $\tilde{x}$, define $\tilde{e} = \tilde{x} - x$ and suppose $c = \mathbb{E}[\tilde{e}\tilde{x}] > 0$. Consider

$$\tilde{x}_{\text{new}} = (1 - \alpha)\tilde{x}, \qquad \tilde{e}_{\text{new}} = \tilde{e} - \alpha\tilde{x}.$$

Then

$$\mathbb{E}[\tilde{e}^2_{\text{new}}] = \mathbb{E}[\tilde{e}^2] - 2\alpha c + \alpha^2 \mathbb{E}[\tilde{x}^2].$$

A sufficiently small $\alpha > 0$ reduces the mean-square error. The best adjustment in this direction is $\alpha = c/\mathbb{E}[\tilde{x}^2]$. Thus a nonzero cross moment reveals a possible improvement. With the opposite error convention $x - \tilde{x}$, the sign of the improving adjustment reverses.

## 6.3 A necessary condition is not a complete optimality test

The identity $\mathbb{E}[e\hat{x}^T] = 0$ alone is not sufficient for MMSE optimality: the constant estimate $\hat{x} = 0$ satisfies it regardless of its accuracy. The full condition is orthogonality to *every* square-integrable function of the available measurement, or equivalently $\mathbb{E}[e \mid z] = 0$.

This stronger condition makes the optimality statement precise. If we add any correction $h(z)$ to the conditional-mean estimate, its error is $e + h(z)$, and

$$\mathbb{E}[\|e + h(z)\|^2] = \mathbb{E}[\|e\|^2] + \mathbb{E}[\|h(z)\|^2] \geq \mathbb{E}[\|e\|^2].$$

# 7 MAP and MMSE interpretations

## 7.1 Maximum a posteriori estimation

After observing $z$, the maximum a posteriori estimate chooses the most likely parameter value according to the posterior density:

$$\hat{x}_{\text{MAP}} = \arg\max_a \, p_{x|z}(a \mid z).$$

For a nonsingular Gaussian posterior $\mathcal{N}(\hat{x}, P)$,

$$p_{x|z}(a \mid z) \propto \exp\left[-\tfrac{1}{2}(a-\hat{x})^T P^{-1}(a-\hat{x})\right].$$

The exponent is maximized at $a = \hat{x}$. Therefore the Gaussian posterior mode and mean coincide:

$$\boxed{\hat{x}_{\text{MAP}} = \hat{x}.}$$

A density maximum is not a positive point probability; for a continuous distribution, any individual point still has probability zero.

## 7.2 Derive the minimum mean-square error estimate

For a candidate value $a$ after observing $z$, minimize the conditional risk

$$J_z(a) = \mathbb{E}[\|x-a\|^2 \mid z] = \int_{\mathbb{R}^n} (x-a)^T(x-a) p_{x|z}(x \mid z)\, dx.$$

Expanding the quadratic,

$$J_z(a) = \mathbb{E}[x^T x \mid z] - 2a^T \mathbb{E}[x \mid z] + a^T a.$$

Thus

$$\nabla_a J_z = -2\mathbb{E}[x \mid z] + 2a, \qquad \nabla_a^2 J_z = 2I \succ 0.$$

The unique minimizing value is

$$\boxed{\hat{x}_{\text{MMSE}} = \mathbb{E}[x \mid z].}$$

Minimizing the conditional risk for each $z$ also minimizes the unconditional risk, because $\mathbb{E}[\|x-g(z)\|^2] = \mathbb{E}[J_z(g(z))]$.

## 7.3 What requires Gaussianity?

The conditional mean is the MMSE estimate for *any* distribution with finite second moments. Gaussianity makes the conditional mean affine in the data and makes it equal to the MAP estimate. For non-Gaussian posteriors, the mean and mode generally differ.

With only second-moment assumptions and independent zero-mean measurement noise, the same gain $K = MH^T(HMH^T+R)^{-1}$ gives the best *affine* estimate. It need not be the unrestricted MMSE estimate. For a nonsingular Gaussian posterior, all these formulas agree:

$$\boxed{\hat{x}_{\text{MMSE}} = \hat{x}_{\text{MAP}} = \bar{x} + MH^T(HMH^T+R)^{-1}(z - H\bar{x}).}$$

# 8 Example: the position of a one-dimensional vehicle

## 8.1 Prior, likelihood, and posterior

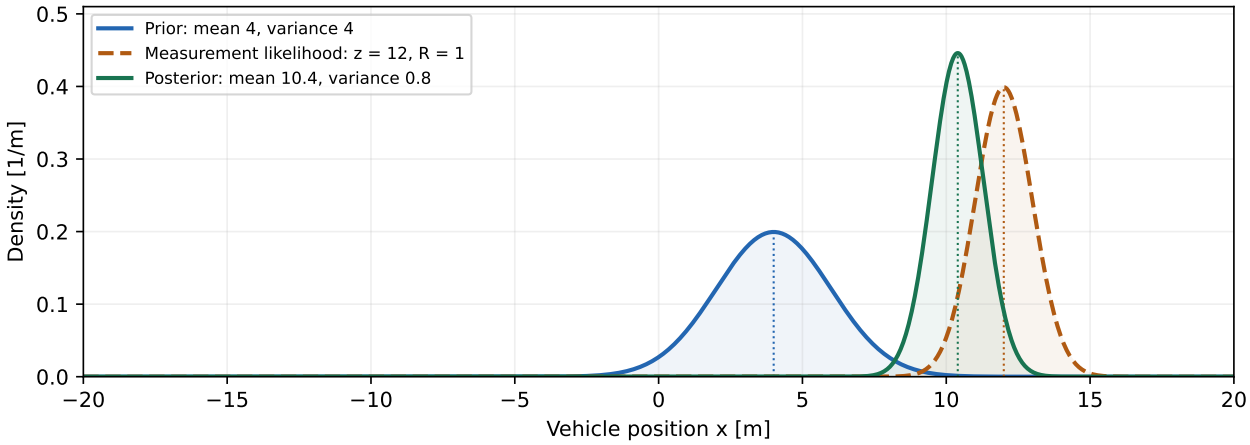
Suppose $x$ is the vehicle's position in meters, with

$$x \sim \mathcal{N}(4, 4), \qquad z = x + v, \qquad v \sim \mathcal{N}(0, 1),$$

and we observe $z = 12$ m. The prior standard deviation is 2 m; the sensor-noise standard deviation is 1 m. The update is

$$K = \frac{4}{4+1} = 0.8, \qquad \boxed{\hat{x} = 4 + 0.8(12 - 4) = 10.4 \text{ m}, \qquad P = 0.8 \text{ m}^2.}$$

Thus $x \mid z = 12 \sim \mathcal{N}(10.4, 0.8)$. The posterior standard deviation is $\sqrt{0.8} \simeq 0.894$ m, smaller than either the prior or sensor standard deviation.




The measurement curve is $p(z \mid x)$ with the observed $z$ held fixed, viewed as a function of candidate position $x$. For this sensor model, it has the shape $\mathcal{N}(z, R)$ and integrates to one over $x$. It is a *likelihood*, not the marginal distribution of future measurements, which is $z \sim \mathcal{N}(\bar{x}, M + R)$ before observing the sensor. Bayes' rule gives $p(x \mid z) \propto p(z \mid x)p(x)$.

## 8.2 Explore the interactive demonstration

**Open the interactive position-estimation demo.** The four sliders set $\bar{x}, M, z, R$; checkboxes show or hide the prior, measurement likelihood, and posterior. The initial values match the example above, and the horizontal axis is fixed at $[-20, 20]$ m.

- Hold $M, R$ fixed and change $z$: the posterior mean moves, but $P$ does not change.
- Set $M = R$: the estimate is halfway between $\bar{x}$ and $z$ and $P = M/2$.
- Decrease $R$ to favor the sensor, or decrease $M$ to favor the prior.

# 9 Summary, practice, and the next lecture

## 9.1 The static Gaussian update

| Quantity | Formula |
|---|---|
| Prior | $x \sim \mathcal{N}(\bar{x}, M)$ |
| Independent measurement noise | $z = Hx + v$, $v \sim \mathcal{N}(0, R)$ |
| Residual and its covariance | $\nu = z - H\bar{x}$, $S = HMH^T + R$ |
| Gain | $K = MH^T S^{-1}$ |
| Posterior mean | $\hat{x} = \bar{x} + K\nu$ |
| Posterior covariance | $P = (I - KH)M(I - KH)^T + KRK^T$ |
| Error convention | $e = \hat{x} - x$ |
| Mean error and orthogonality | $\mathbb{E}[e] = 0$, $\mathbb{E}[e \mid z] = 0$, $\mathbb{E}[e\hat{x}^T] = 0$ |

## 9.2 Practice

1. Two independent fair dice have sum $X$; the first die is $Z$. Find $\mathbb{E}[X]$, $\mathbb{E}[X \mid Z = 3]$, and $\mathbb{P}(X = 7 \mid Z = 3)$.
2. If $X \sim \mathcal{N}(4, 4)$, identify its standard deviation and the probability that $2 \leq X \leq 6$.
3. Use $\bar{x} = 4$, $M = 4$, $z = 12$, $R = 4$ to find $K, \hat{x}, P$. Compare with Section 8.
4. Why can changing $z$ change the posterior mean without changing the posterior covariance in the linear Gaussian model?
5. Does $\mathbb{E}[e] = 0$ mean every realized error is zero? Does $\mathbb{E}[e\hat{x}^T] = 0$ by itself prove MMSE optimality?

## 9.3 Brief answers

1. $\mathbb{E}[X] = 7$, $\mathbb{E}[X \mid Z = 3] = 3 + 3.5 = 6.5$, and $\mathbb{P}(X = 7 \mid Z = 3) = 1/6$.
2. The standard deviation is 2; the interval is one standard deviation on either side of the mean, with probability approximately 0.6827.
3. $K = 1/2$, $\hat{x} = 8$ m, $P = 2$ m$^2$. The noisier measurement receives less weight and leaves more uncertainty.
4. The posterior covariance depends on $M, H, R$; the observed residual changes the mean through $K(z - H\bar{x})$.
5. No to both. Zero mean is an average statement. MMSE optimality requires orthogonality to every function of the available data, not just to the estimate itself.

## 9.4 Week 13: Kalman filtering and LQG

The next lecture introduces dynamic stochastic state models, prediction and correction, filter covariance and Riccati equations, and innovations. It then combines Kalman estimation with LQR through the separation principle to obtain LQG output feedback.

**Source:** Handwritten MAE 6246 lecture, November 13, 2026: probability and Gaussian properties (pp. 1–3), parameter estimation (pp. 4–5), and orthogonality, MAP, and MMSE (pp. 5–6). The vehicle example matches the course demonstration.